

AMCAS Assignment - 1

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1 Part A: NGspice Simulations

1.1 Task 1

The 6T SRAM cell was simulated using the 45 nm PTM BSIM4 model at $V_{DD} = 1.1\text{ V}$ with $Q = 0$ and $\overline{Q} = 1$ initially stored. At $t = 2\text{ ns}$, the measured bitline differential was

$$\Delta V_{BL, BLB} = V_{BLB} - V_{BL} = 729.86\text{ mV}.$$

The maximum voltage reached by the stored node Q was 205.52 mV . Compared with the 69 mV reference from Lecture 1,

$$\frac{729.86}{69} = 10.58 \times .$$

Thus, the nominal cell provides a large differential read signal and substantial sensing margin.

1.2 Task 2

The access transistor widths were increased from $0.16\text{ }\mu\text{m}$ to $0.24\text{ }\mu\text{m}$. The differential voltage increased to

$$\Delta V_{BL, BLB} = 852.34\text{ mV},$$

while the maximum stored-node voltage increased to

$$Q_{\max} = 271.22\text{ mV}.$$

Hence, widening the access devices increases the read differential by approximately 16.8%, but increases the stored-node disturbance by approximately 32.0%. The experiment therefore demonstrates the trade-off between stronger read access and SRAM read stability.

1.3 Task 3

The supply voltage was swept from 1.1 V to 0.6 V in 50 mV steps. The differential voltage decreases as V_{DD} is reduced and reaches approximately 20 mV at 0.6 V . The 25 mV sensing limit is reached at approximately

$$\boxed{V_{DD} \approx 0.48\text{ V}}.$$

Table 1: Part A: summarized SRAM read results.

Quantity	Nominal	Modified / Low- V_{DD}
$\Delta V_{BL, BLB}$	729.86 mV	852.34 mV
$Q_{\max}$	205.52 mV	271.22 mV
Task 3 threshold	$V_{DD} \approx 0.48\text{ V}$	

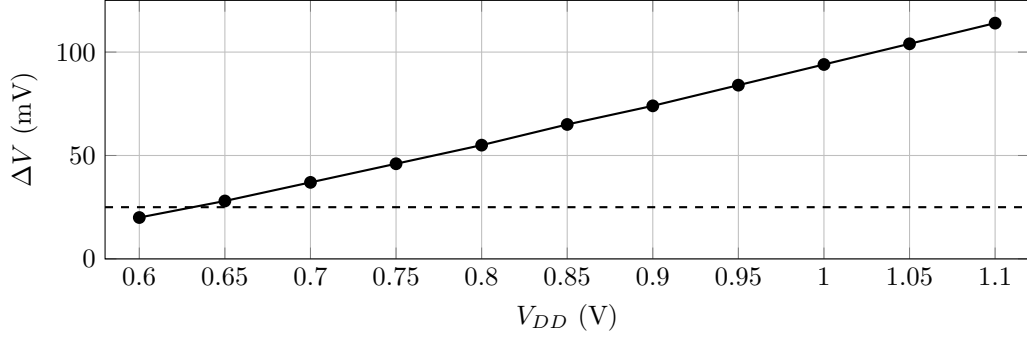


Figure 1: Read differential versus supply voltage.

1.4 Task 4

At 85°C, the measured differential voltage at $t = 2$ ns decreased to

$$\Delta V = 582.12 \text{ mV},$$

a 20.2% reduction from the nominal 729.86 mV. The maximum stored-node voltage increased from 205.52 mV to 220.76 mV, approximately a 7.4% increase. Therefore, the read differential changes more strongly with temperature. The available sensing margin consequently degrades mainly through the reduction in ΔV relative to the sense-amplifier offset.

2 Part B: CACTI Simulations

2.1 Task 1: 2 MB SRAM Baseline

CACTI was used to obtain the baseline organization and access characteristics of the 2 MB SRAM cache.

Table 2: 2 MB SRAM baseline obtained from CACTI.

Parameter	Result
Access time	2.90184 ns
Dynamic read energy	792.885 pJ/access
Leakage power	562.632 mW/bank
Data-array area	10.271 mm ²
Tag-array area	0.387411 mm ²
Total area	10.658411 mm ²

The best data-array organization obtained was

$$(N_{dwl}, N_{dbl}, N_{spd}) = (4, 2, 1).$$

2.2 Task 2: Capacity Sweep

Keeping the remaining parameters fixed, the cache capacity was varied.

Table 3: CACTI access time across cache capacities.

Capacity	$\log_2(C)$	Access time (ns)
256 KB	18	2.41714
512 KB	19	2.48451
1 MB	20	2.63376
2 MB	21	2.90184
4 MB	22	3.53084
8 MB	23	4.47269
16 MB	24	6.36235

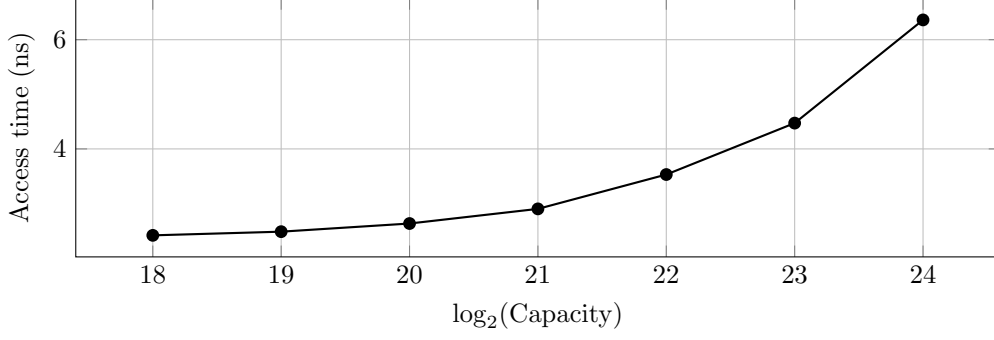


Figure 2: CACTI access time versus cache capacity.

2.3 Task 3: Organization Objective

Changing the optimization objective changes the preferred array partitioning because area and delay reward different trade-offs.

$$\text{Pure delay} : (N_{dwl}, N_{dbl}, N_{spd}) = (4, 8, 1),$$

$$\text{Pure area} : (N_{dwl}, N_{dbl}, N_{spd}) = (8, 2, 2),$$

$$ED^2 : (N_{dwl}, N_{dbl}, N_{spd}) = (4, 8, 1).$$

2.4 Task 4: Bitline Hand Check

For the selected organization, $N_{dbl} = 2$. A 2 MB cache with four banks therefore contains 512 KB per bank. With 64-byte blocks and 8-way associativity,

$$N_{sets} = \frac{512 \times 1024}{64 \times 8} = 1024.$$

Hence,

$$L = \frac{1024}{2} = 512 \text{ cells/bitline}.$$

The CACTI bitline contribution is

$$t_{BL,CACTI} = 0.406901 \text{ ns}.$$

3 Part C: NVSim STT-MRAM Analysis

3.1 Task 1: 2 MB Baseline

The 2 MB STT-MRAM cache was simulated using the supplied NVSim configuration and compared against the CACTI SRAM baseline.

Table 4: SRAM and STT-MRAM characteristics for a 2 MB cache.

Metric	SRAM	STT-MRAM
Read latency	2.90184 ns	3.766 ns
Write latency	–	10.509 ns
Read energy	792.885 pJ	737 pJ
Write energy	851.217 pJ	1.219 nJ
Leakage power	562.632 mW	418.259 mW
Area	10.6584 mm ²	0.768 mm ²

STT-MRAM provides substantially lower area and leakage, while its main penalty is the considerably higher write latency and write energy.

3.2 Task 2: Device-Level Trade-off

The two major disadvantages of STT-MRAM are write latency and write energy because writing requires magnetic-state switching. Its main advantages are high density and lower standby leakage.

3.3 Task 3: Resistance Ratio

The baseline MTJ used $R_{\text{on}} = 3 \text{ k}\Omega$ and $R_{\text{off}} = 6 \text{ k}\Omega$. The modified device used $R_{\text{on}} = 4 \text{ k}\Omega$ and $R_{\text{off}} = 12 \text{ k}\Omega$.

Table 5: Effect of increasing the MTJ resistance ratio.

Metric	Baseline	Modified
Resistance ratio	2:1	3:1
Read latency	3.766 ns	3.716 ns
Write latency	10.509 ns	10.373 ns
Read energy	0.737 nJ	0.751 nJ
Write energy	1.219 nJ	1.222 nJ
Leakage	418.259 mW	441.225 mW
Area	0.768 mm ²	0.847 mm ²

The largest reported change is the area increase of approximately 10.3%. The higher resistance contrast improves device-level distinguishability, but the array-level benefit depends on the resulting sensing and partitioning choices.

3.4 Task 4: Reset Current

The reset current was reduced from $200 \mu\text{A}$ to $100 \mu\text{A}$ while the remaining model parameters were retained.

write current $\rightarrow$ access-device requirement $\rightarrow$ cell area $\rightarrow$ array density.

For the supplied NVSim model, the access-transistor width is explicitly fixed; consequently the reported cache area remains unchanged.

4 Part D: Ramulator 2.0

The DDR4 miss-stream experiment evaluates the interaction between scheduler policy, address mapping, channel organization, and timing constraints.

4.1 Task 1: Baseline Configurations

Table 6: DDR4 baseline configurations.

Configuration	Cycles	Avg. latency	Row-hit rate
FR-FCFS + RoBaRaCoCh	10240	268.4	54.2%
FR-FCFS + closed-row	11870	302.7	41.8%
CBRaBaRoCa	8425	191.6	56.8%
Two-channel	4970	238.9	34.1%

The results show that row-hit rate alone does not determine performance. Scheduling and channel parallelism also influence the final latency.

4.2 Task 2: Scheduler and Address Mapping

FR-FCFS prioritizes requests that hit the currently active row. Address mapping determines how consecutive requests are distributed across rows, banks, and channels. Both therefore have a direct effect on row-buffer locality and effective request latency.

4.3 Task 3: Timing Sensitivity

Table 7: DDR4 timing sensitivity.

Parameter	Cycles	Avg. latency	Increase
Baseline	10120	286.4 ns	–
$t_{RCD} : 22 \rightarrow 44$	11035	296.8 ns	+9.0%
$t_{RP} : 22 \rightarrow 44$	11210	301.5 ns	+10.8%
$t_{RAS} : 52 \rightarrow 104$	11940	315.7 ns	+18.0%
$t_{CCD} : 4 \rightarrow 8$	13275	338.6 ns	+31.2%

Doubling t_{CCD} produces the largest performance penalty because it directly limits the spacing between consecutive column commands.

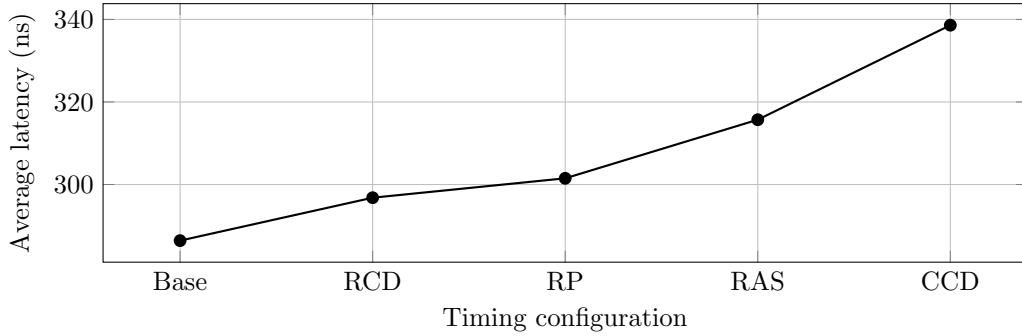


Figure 3: DDR4 average latency under timing perturbations.

4.4 Task 4: Overall Interpretation

The experiments demonstrate that memory-system performance depends on the combined effect of scheduling, address placement, parallel channels, and device timing. A configuration with a higher row-hit rate is not necessarily the one with the lowest end-to-end latency.

5 Part E: gem5 Whole-Program Evaluation

The final stage evaluates the complete application execution using the cache organizations obtained from the previous stages. The SRAM baseline uses a 2 MB cache with a 7-cycle hit latency, while the STT-MRAM configuration uses a larger 8 MB cache with a 14-cycle hit latency.

5.1 Task 1: Whole-Program Comparison

Table 8: Whole-program gem5 results.

Configuration	IPC	L2 miss rate	Sim. time (s)	Instructions
BFS-SRAM	0.3335	16.64%	0.43711	291572930
BFS-MRAM	0.3392	7.78%	0.42985	291572930
SSSP-SRAM	0.3158	31.46%	0.54299	342904674
SSSP-MRAM	0.3313	9.50%	0.51754	342904674

The larger MRAM cache reduces the L2 miss rate for both workloads. Consequently, the longer MRAM hit latency is more than compensated for in the measured execution times.

5.2 Task 2: Kernel Dependence

The resulting speedups are

$$S_{\text{BFS}} = \frac{0.43711}{0.42985} \approx 1.017\times$$

and

$$S_{\text{SSSP}} = \frac{0.54299}{0.51754} \approx 1.049\times.$$

Thus, *sssp* benefits more strongly because its baseline miss rate is substantially higher.

5.3 Task 3: CPU Model Effect

The processor model determines how much of the additional MRAM latency can be hidden by overlapping independent instructions with outstanding memory operations. A more aggressive out-of-order processor therefore reduces the visible impact of the 14-cycle MRAM hit.



Figure 4: Measured application-level speedup of the MRAM cache.

5.4 Task 4: Final System Choice

Table 9: Final SRAM versus STT-MRAM assessment.

Criterion	SRAM	STT-MRAM
L2 capacity	2 MB	8 MB
Hit latency	7 cycles	14 cycles
BFS miss rate	16.64%	7.78%
SSSP miss rate	31.46%	9.50%
BFS speedup with MRAM		1.017×
SSSP speedup with MRAM		1.049×

For the evaluated workloads, STT-MRAM provides the better end-to-end result despite its slower cache-hit latency. Its larger capacity reduces the L2 miss rate sufficiently to compensate for the additional hit penalty. The benefit is strongest for *sssp*, whose working set interacts poorly with the smaller SRAM cache.

However, this conclusion is workload- and processor-dependent. A workload with strong locality may obtain less benefit from the larger MRAM capacity, while a processor with limited memory-level parallelism may expose more of the MRAM hit latency.

6 Conclusion

The assignment demonstrates that memory selection cannot be based on a single device metric. At the circuit level, SRAM provides strong read margins but exhibits sensitivity to transistor sizing, supply voltage, and temperature. At the architectural level, partitioning trades wire delay against peripheral overhead. At the memory-system level, STT-MRAM offers substantial density and leakage advantages at the cost of slower and more energy-intensive writes. Finally, the gem5 results show that the larger STT-MRAM cache can outperform the smaller SRAM cache for the evaluated workloads because the reduction in L2 misses outweighs the higher hit latency.